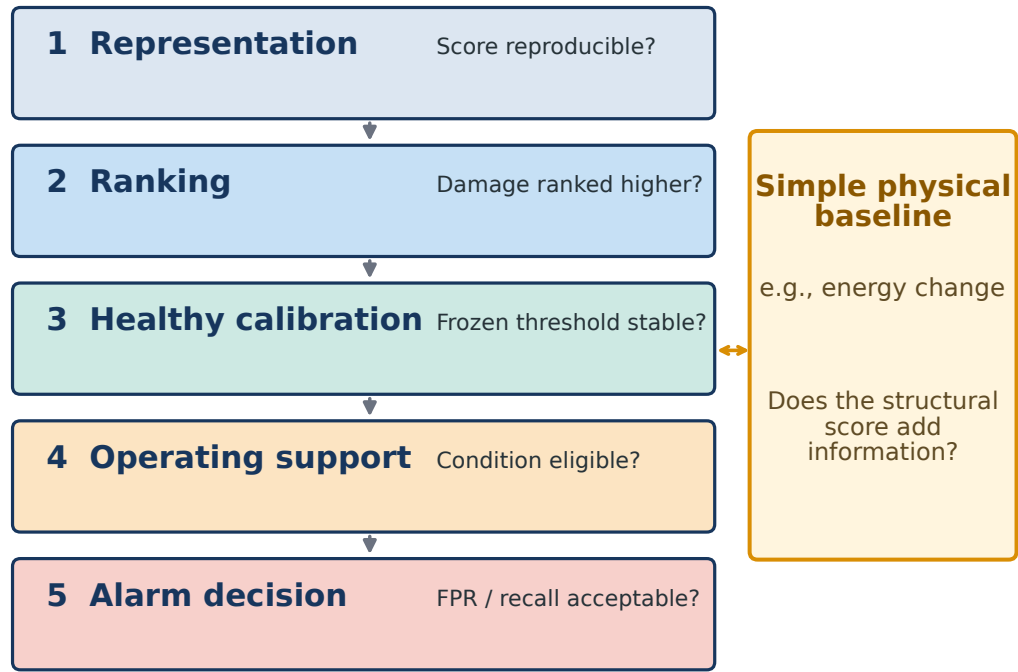


High AUROC ≠ reliable alarms

Qualification is a chain: failure at any required layer limits the claim.



Output: score + calibration + support + alarm / abstain

External COPV confirmation

Structural macro AUROC **0.9890**

Observed healthy alarms **0 / 39**

Supported damage recall **67/132 = 0.5076**

Damage support **132/168 = 0.7857**

Structural - energy AUROC **-0.0065**

Mandatory gates **7 / 13 passed**

STRICT OUTCOME: FAIL